CIVILIZATION

THE ABSOLUTE TRUTH.

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Venture Overview

MISSION STATEMENT :

To accelerate humanity's advancement on the Kardashev Scale by pioneering breakthrough technologies that redefine the future of Human Civilization.

OVERVIEW :

CIVILIZATION is a deep-tech parent company driving humanity's advancement on the Kardashev Civilizational Scale. Incorporated in November 2024, it hosts six subsidiaries tackling frontier industries: **ETERNAL** (nuclear space engines), **JOULE** (India's first fusion reactor), **DEEP** (underwater hyperspectral imaging & advanced ROVs), **PLANCK** (error-proof qubits & quantum computing), **GENETICS** (WBE), and **BEYOND AI** (AGI trained on quantum systems). **Currently at TRL 2–3, with IIT collaborations and secured trademarks, CIVILIZATION is uniquely positioned to pioneer breakthrough technologies. Its mission is singular yet expansive—accelerating human civilization's technological trajectory while establishing India as a global leader in deep-tech innovation.**

What Is Kardashev Scale?

Type 0 Civilization

Where humanity is today (around 0.73). We rely on fossil fuels, nuclear power, and emerging renewables but don't fully harness Earth's energy.

Type I Civilization

A planetary civilization. Uses **100% of Earth's energy resources**: solar, wind, geothermal, oceans, etc. Full control of the planet's energy.

Type II Civilization

A stellar civilization. Harnesses the **entire energy of the Sun.**

Type III Civilization

A galactic civilization. Controls energy of **all stars in its galaxy**.

Type IV Civilization

A universal civilization. Utilizes energy from **the entire universe.**

Type V Civilization

A multi-versal civilization. Harnesses energy across **multiple universes**.

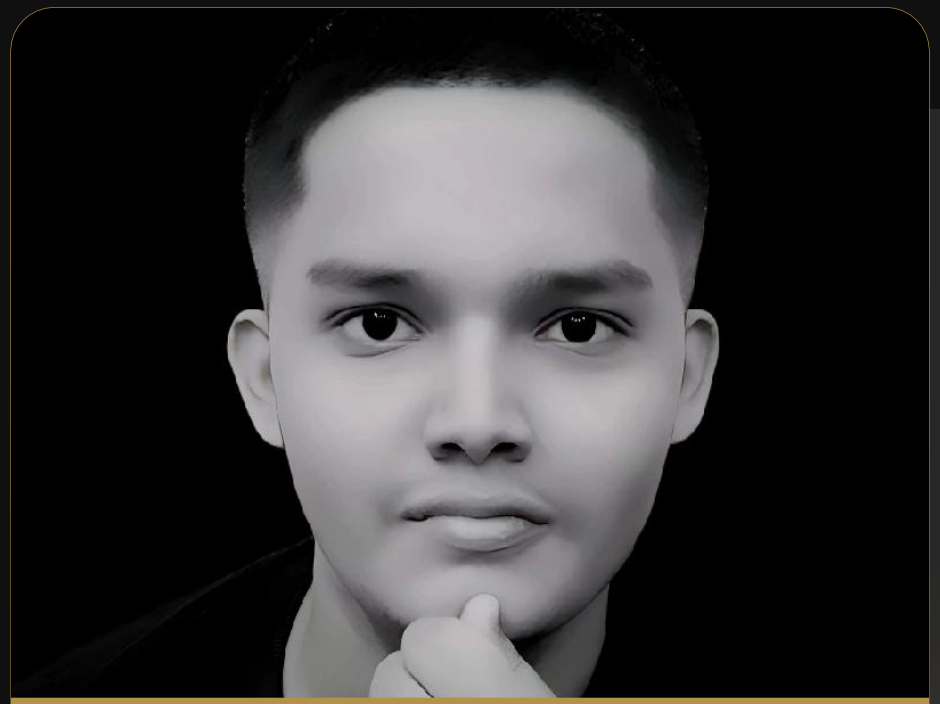
Type VI Civilization

Masters control over **space-time itself**, manipulating dimensions, black holes, and cosmic structures.

Type VII Civilization

Civilization becomes indistinguishable from **cosmic consciousness or god-like entities**, mastering all aspects of existence and creation.

Founder's Profile

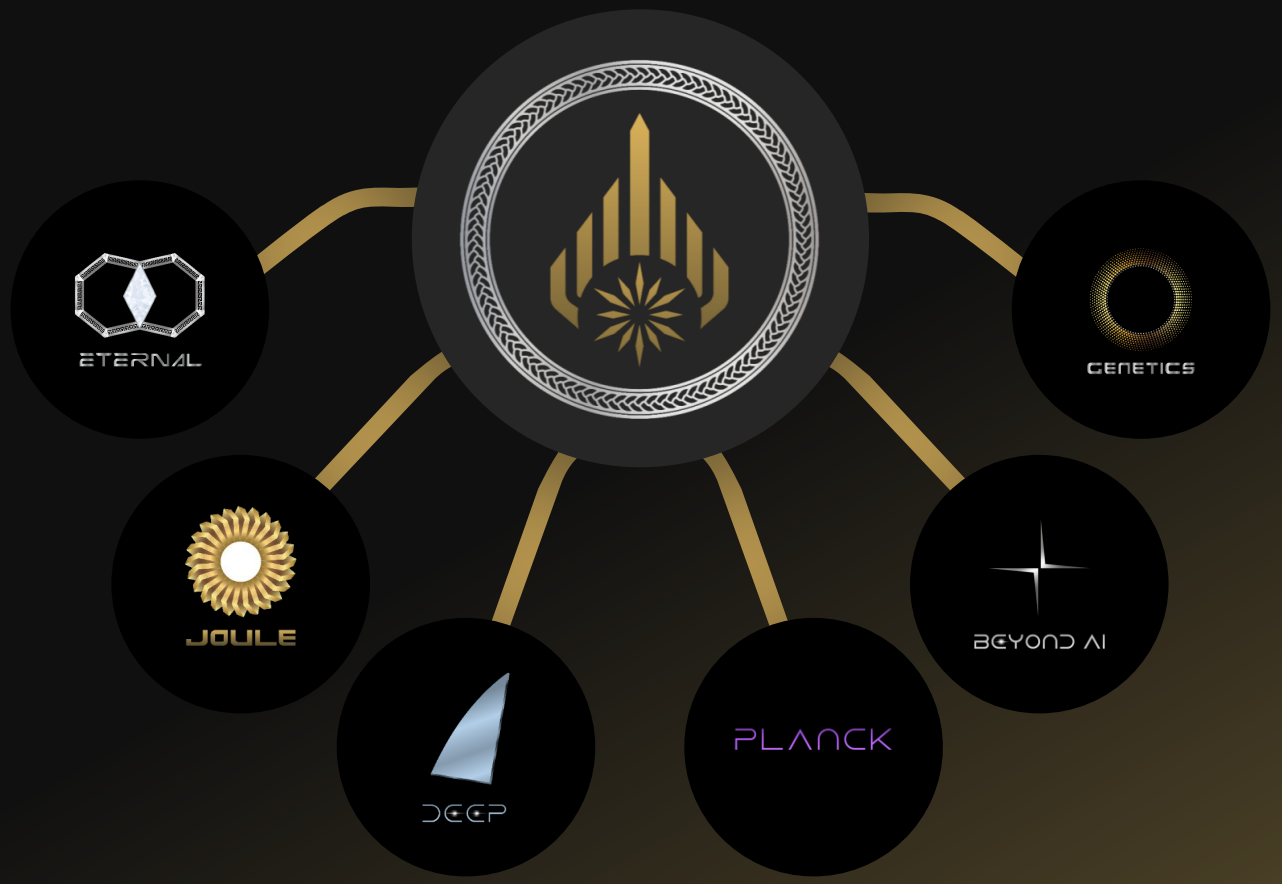


Surya Swarup

Founder & Ceo

Pursuing a **Bachelor's in Business Administration at IIM Bangalore** as a First-year student, I bring a strong foundation in academics and leadership. My academic journey includes achieving 94.2% in Class 12 (School Rank 1) and 95.2% in Class 10, reflecting consistent excellence. I am a **National Gold Medalist in Commerce Olympiad** and **Silver Medalist in English Olympiad.** Additionally, I had the privilege of being recognized as a learner at ISRO, which further broadened my perspective on **innovation and research.**

OUR SUBSIDIARIES



ETERNAL

Problem Statement

Space exploration is currently limited by propulsion technology. Traditional chemical rockets are bulky, inefficient, and environmentally damaging. For example, launching a single Falcon 9 rocket releases **over 330 tons of CO_2 per launch,** comparable to the annual emissions of dozens of cars. Moreover, chemical propulsion cannot deliver the thrust-to-weight ratio necessary for rapid interplanetary travel.

These inefficiencies make long-duration missions to Mars and beyond extremely time-consuming **(taking 6–9 months to reach Mars)** and resource-intensive. Current propulsion models also severely limit payload capacity, which is a critical bottleneck for deep space missions, satellite deployment, and future colonization efforts. **Without breakthrough propulsion, humanity cannot scale space access affordably or sustainably.**

Solution Statement

Current Stage: Idea/Conceptual Design (TRL 2–3)

Focus Area:

At this stage, **ETERNAL** is not targeting full rocket development but concentrating on the **most crucial subsystem of space exploration — the propulsion engine.**

Why Nuclear Propulsion?

- Higher Thrust-to-Weight Ratio
- Reduced Mass & Cost
- Eco-Friendly Alternative
- Faster Travel Times

Initial Mission:

Develop a **nuclear-capable rocket engine** that overcomes the fundamental limitations of chemical propulsion.

Vision:

Position ETERNAL as a **specialized propulsion leader** in the space industry, with nuclear-capable engines enabling interplanetary travel, satellite logistics, and deep-space missions.

JOULE

Problem Statement

India, like the rest of the world, continues to depend heavily on fossil fuels, which account for nearly **70% of its electricity production.** While solar and wind energy are growing, they are intermittent, land-intensive, and dependent on weather patterns, making them unreliable as a base-load power source. This leads to grid instability and a continued reliance on coal and gas.

If India remains dependent on fossil fuels, its energy demand—which is expected to **double by 2040**—will push it into a severe climate and resource crisis.

JOULE

Solution Statement

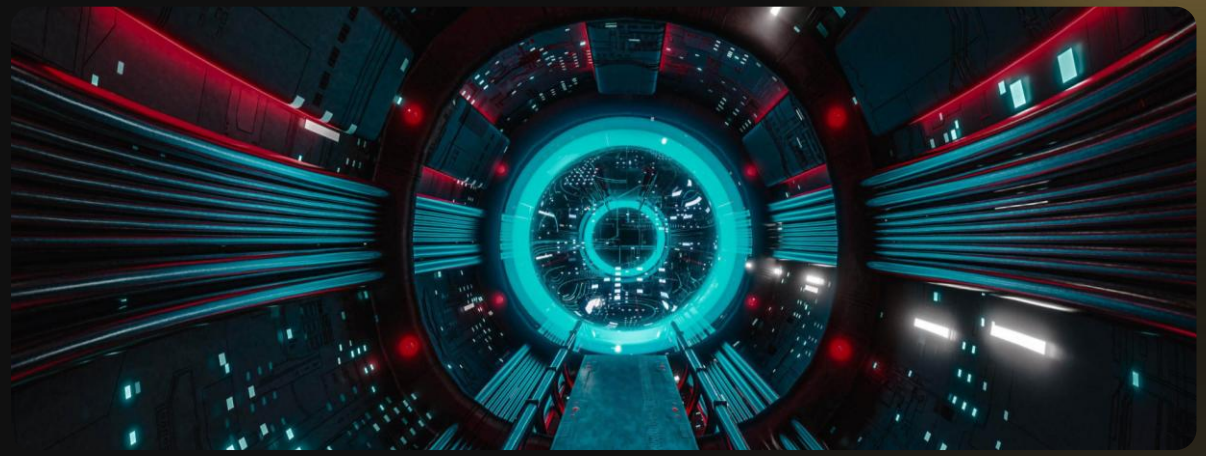
Current Stage: Conceptual R&D (TRL 2–3)

Focus Area:

At the initial phase, **JOULE** is dedicated to building **India's first nuclear fusion reactor** — a groundbreaking step in achieving limitless, clean, and sustainable energy.

Why Fusion?

- **Limitless Energy Source**
- **Zero Greenhouse Emissions**
- **No Long-Lived Nuclear Waste**
- **Energy Density:** 1 kg of fusion fuel releases energy ≈ **10 million kg of fossil fuel**.
- **Safety:** No risk of runaway reactions.



Strategic Significance – First in India

- India is already part of the **ITER project (France),** contributing materials and R&D, but **no indigenous fusion reactor exists yet.**
- Building the first Indian fusion reactor positions JOULE as a national pioneer in next-gen clean energy, with both domestic and global market potential.

Vision:

To make **India's first fusion reactor operational**, giving the country **energy independence** & positioning it at the forefront of the **global clean energy revolution.**

Problem Statement

The ocean covers over **70% of Earth's surface, yet more than 80% remains unexplored.** Critical ecosystems, mineral resources, and biodiversity are hidden beneath, but our current tools are insufficient. Traditional sonar and imaging systems fail to capture fine details, while most ROVs and AUVs are costly, limited in endurance, and unsuitable for high-resolution, deep-sea exploration.

This knowledge gap has real-world consequences: from **undiscovered marine life to untapped mineral reserves** and even the understanding of climate systems regulated by ocean currents. Without advanced imaging and robotics, humanity risks missing out on vital resources and insights that could drive medicine, climate science, and sustainable development.

Solution Statement

Current Stage: Conceptual R&D / Early Prototyping (TRL 2–3)

Focus Area:

In its initial phase, **DEEP** is focused on developing:

- **Hyperspectral Imaging Systems** for underwater exploration.
- **Highly advanced ROVs (Remotely Operated Vehicles) and AUVs (Autonomous Underwater Vehicles).**

Significance of Hyperspectral Imaging Underwater:

- Provides detailed spectral information beyond what traditional cameras or sonar can capture.
- Enables identification of minerals, corals, marine organisms, pollutants, and archaeological sites with high accuracy.
- Critical for seabed mapping, resource discovery, and environmental monitoring.

Significance of Advanced ROVs and AUVs:

- Expand exploration beyond diver limits — deep-sea zones >6,000 meters.
- Enable real-time data collection for climate science, marine biology, and resource mapping.
- Applications in offshore energy, submarine cable inspections, undersea mining, and defense.

Strategic Opportunity:

- Oceans cover ~71% of Earth's surface, yet >80% remains unexplored (NOAA).
- India, with a coastline of ~7,500 km, is investing in the Deep Ocean Mission (₹4,077 crore / $500M, 2021–2026).

Vision:

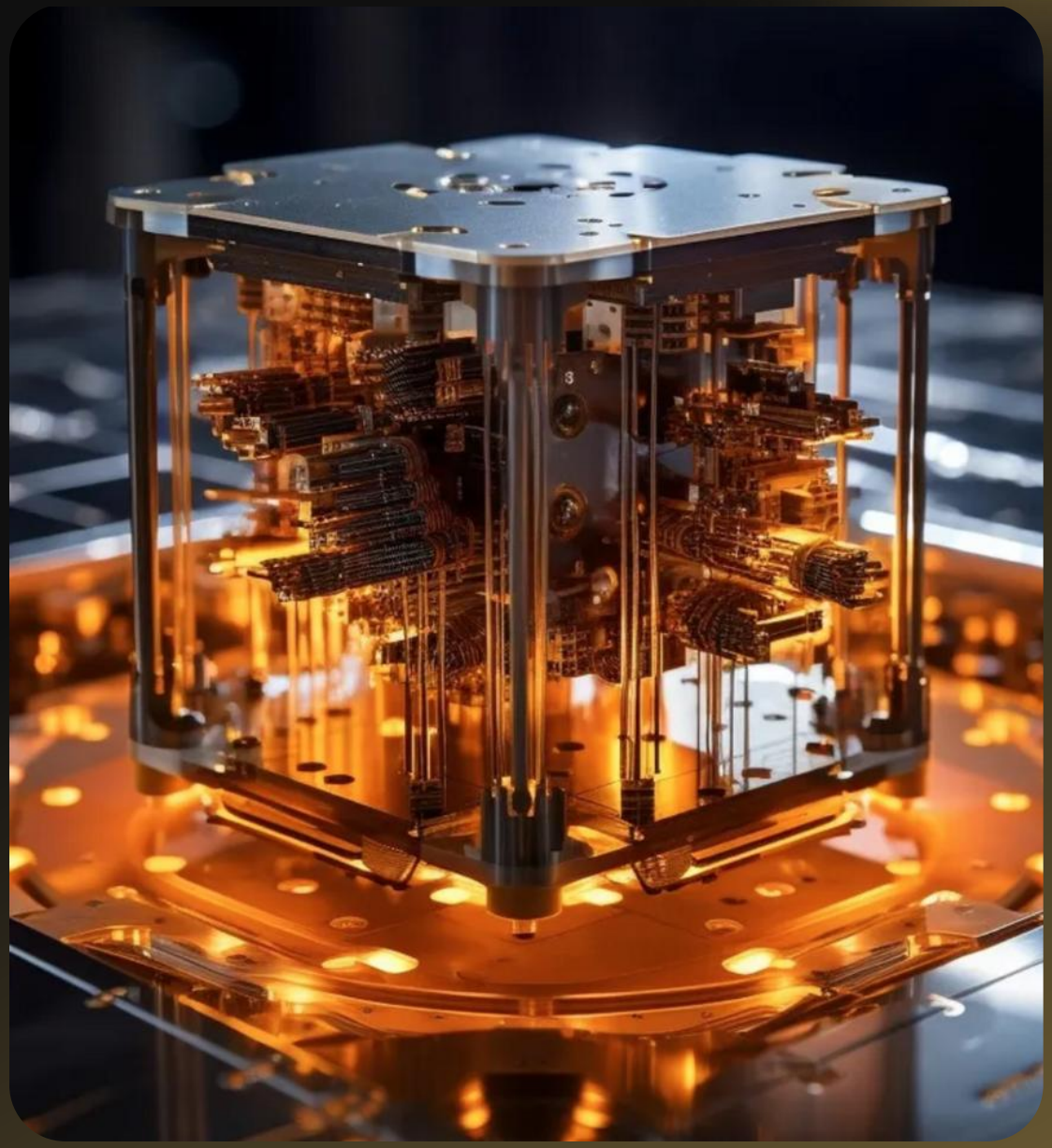
To enable India to unlock the blue economy (estimated at $1 trillion by 2035) by providing world-class underwater exploration technologies.

PLANCK

Problem Statement

Quantum computing promises exponential leaps in computational power, but today's prototypes are impractical. They rely on fragile qubits that must be kept at temperatures close to **absolute zero (-273ºC)**. These systems are error-prone and unstable, making them unsuitable for real-world applications outside of labs.

As a result, despite billions of dollars in global investment, quantum computing remains stuck in early research stages. Without solving the **error-correction and operability at normal conditions** challenge, the technology cannot transition into usable platforms for industries like pharmaceuticals, materials discovery, and artificial intelligence. This bottleneck delays breakthroughs that could reshape economies.



PLANCK

Solution Statement

Current Stage: Conceptual R&D (TRL 1–2)

Focus Area:

PLANCK is working on developing the **world's first error-proof qubit** capable of operating at **normal temperature and pressure**.

- Current quantum computers require **cryogenic cooling (~15 mK)** and ultra-controlled conditions.
- Error correction is the biggest bottleneck: today's qubits have **error rates between 0.1–1% per gate operation** (Google, IBM).
- By solving stability and scalability at ambient conditions, PLANCK could unlock **commercially viable quantum computing**.

Significance:

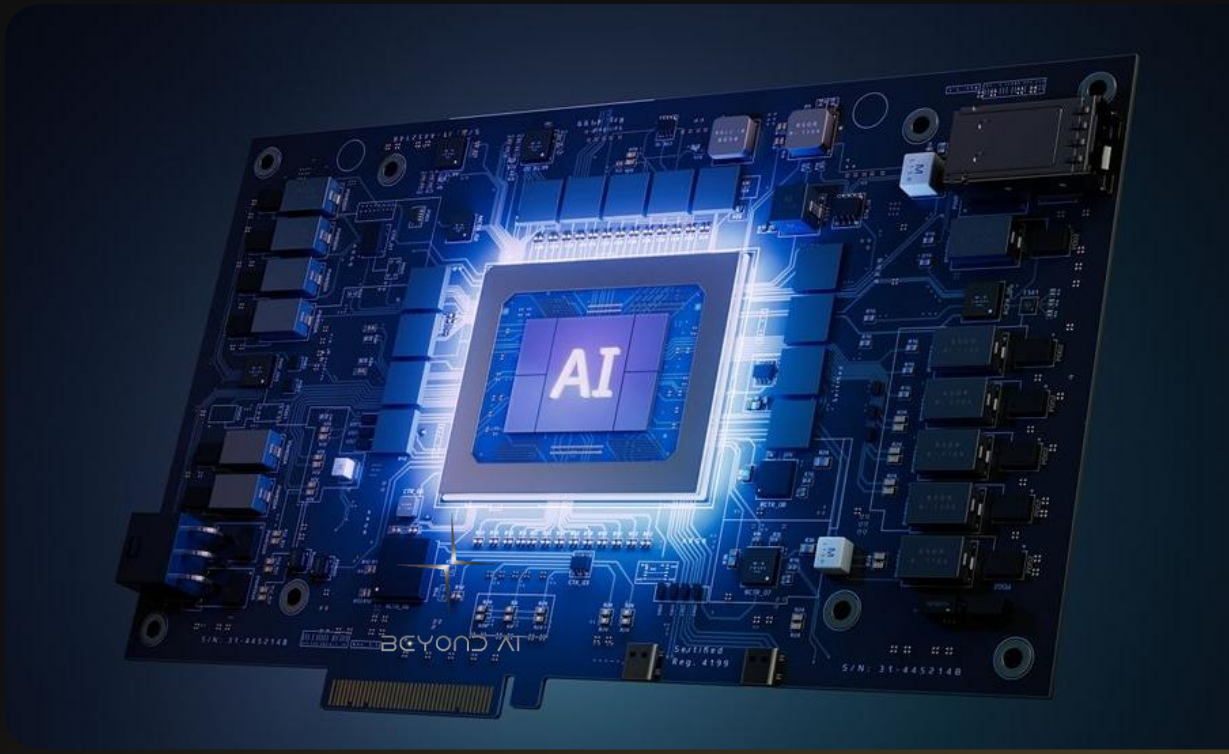
- Quantum computers could solve problems **trillions of times faster** than classical supercomputers for drug discovery, materials science, finance, climate modeling, and AI training.
- The global **quantum computing market** is projected to grow from $**1.3B in 2024 to $5.3B by 2029.**

BEYOND AI

Problem Statement

Current AI training relies heavily on **GPU-based computation**, which faces **scalability, energy, and cost bottlenecks**. Training frontier AI models already requires **millions of GPU hours**, leading to rising energy consumption, supply shortages, and prohibitive costs.

However, **GPUs are not fundamentally designed for quantum-level parallelism** required by Artificial General Intelligence (AGI). Quantum computing promises exponential speed-ups for certain tasks such as optimization, machine learning, and data simulation, but existing quantum **hardware struggles with fragile, error-prone qubits and extreme cooling requirements**.



Thus, the bottleneck is clear: **without a paradigm shift from GPUs to stable, error-resistant quantum computing, the pursuit of AGI will remain constrained by today's hardware limits.** BEYOND AI directly addresses this by leveraging PLANCK's work on stable qubits to unlock scalable, energy-efficient quantum AI training.

BEYOND AI

Solution Statement

Current Stage: Early Research & Integration (TRL 2–3)

Focus Area:

BEYOND AI aims to build Artificial General **Intelligence (AGI)** — AI capable of human-level reasoning and adaptability.

- The challenge: training **AGI requires computational power** far beyond today's GPUs/TPUs.
- PLANCK's quantum computing breakthroughs will serve as the **computational backbone** for Beyond AI's AGI development.

Significance:

- The **global AI market** is projected to exceed **$1.8 trillion by 2030 (PwC/Statista).**
- Current large language models cost **tens of millions of dollars** to train (e.g., GPT-4 training estimated at **$63M–$100M** on GPUs). Quantum acceleration could reduce both **time and cost** dramatically.
- AGI trained on quantum systems could tackle **multi-domain reasoning,** ethical decision-making, and real-time global simulations — beyond current narrow AI.

Complementarity
(PLANCK + BEYOND AI)

PLANCK → Hardware Breakthrough: Error-proof, room-temperature qubits → scalable quantum computer.

BEYOND AI → Software Leap: AGI that leverages PLANCK's quantum platform for training and reasoning.

Together: A hardware-software symbiosis, enabling a quantum-trained AGI — a step-change beyond both today's quantum startups and AI labs.

GENETICS

Problem Statement

The human brain, with **86 billion neurons and over 100 trillion synapses**, remains beyond full scientific replication. Current AI and neuromorphic systems simulate fragments of thought but fail to capture the emergent essence of consciousness.

Genetics seeks to change this through **Whole Brain Emulation (WBE)** — digitally reconstructing and simulating the human mind using neuroimaging, connectomics, and hybrid neuromorphic–quantum computation. The goal: to achieve continuous, conscious, and self-sustaining digital existence — advancing humanity toward the frontier of **digital immortality**.

GENETICS

Solution Statement

Current Stage: Conceptual R&D (TRL 1–2)

Focus Area:

Genetics focuses on developing a **Whole Brain Emulation (WBE)** framework — digitally reconstructing and simulating the human brain using advances in connectomics, neuroimaging, and hybrid neuromorphic–quantum computation.

Market Significance:

- Establishes the foundation for the **digital immortality industry**, projected to redefine healthcare.
- Enables breakthroughs in **AI, neuroscience, and longevity research.**

Why Neuromorphic Computing?

- Bridges the gap between artificial and biological intelligence.
- Enables long-term cognitive preservation and identity continuity.
- Drives the evolution of self-aware AI systems rooted in human consciousness.
- Marks the next leap in humanity's transition toward digital immortality.

Progress So Far :

Company Incorporation:

CIVILIZATION incorporated on **19th November 2024**

Subsidiary Coverage:

Six companies legally defined under CIVILIZATION:
ETERNAL, JOULE, DEEP, PLANCK, GENETICS, BEYOND AI.

IP Protection:

All **legal trademarks acquired** for parent company and subsidiaries.

Significance of TRL 2–3:

- TRL 2: Technology concept formulated.
- TRL 3: Experimental proof-of-concept.
- Shows that CIVILIZATION is beyond the "idea" stage → already **exploring feasibility, models, and R&D pathways.**

Technology Readiness Level (TRL):

- ETERNAL (Space): TRL 2–3 → Conceptual models of nuclear-capable propulsion engines.
- JOULE (Renewable Energy): TRL 2–3 → Feasibility phase for India's first fusion reactor.
- DEEP (Underwater Exploration): TRL 2–3 → Early design phase for hyperspectral imaging systems & advanced ROV/AUV prototypes.
- PLANCK (Quantum Tech): TRL 2–3 → Theoretical design of error-proof, room-temperature qubits.
- BEYOND AI (AI): TRL 2–3 → Framework for AGI development aligned with PLANCK's quantum computing backbone.
- GENETICS (Biotech): TRL 2–3 → Conceptual phase.

Next 3-Month Roadmap

Objective:

Advance all subsidiaries from **TRL-2** → **TRL-3**, establishing experimental proof-of-concept.

Planned Activities:

- Engage with **peers and professors at IITs** across relevant domains (aerospace, energy, ocean engineering, quantum computing, biotech, AI).
- Co-develop **early-stage experimental models** and validate initial feasibility for each subsidiary's core technology.
- Lay groundwork for **academic collaborations, lab access, and joint publications.**

Why TRL-3 is Significant:

- **Validation Step:** Moving from theoretical models (TRL-2) to experimental proof-of-concept (TRL-3)
- This stage helps identify **technical feasibility, limitations, and scalability** before investing in prototypes.
- Academic collaboration ensures **credible validation** and enhances chances of attracting future R&D funding.
- Positions CIVILIZATION as a **serious, research-backed venture** rather than a purely conceptual initiative.

Competitor Landscape & Differentiation

1 ETERNAL

Global Competitors:

- NASA + DARPA DRACO program → nuclear propulsion demo by 2027.
- Roscosmos → megawatt-class nuclear engines.
- ISRO → exploring early nuclear propulsion concepts.

Differentiator:

ETERNAL focuses exclusively on propulsion engines (not rockets), with the goal of creating lighter, eco-friendly, nuclear-capable systems — a niche underserved in the commercial space sector.

2 JOULE

Global Competitors:

- ITER (France): multinational, world's largest fusion project, "first plasma" expected by 2025.
- Commonwealth Fusion Systems (US): raised $2B+, compact tokamak design.
- China's EAST reactor: achieved 120 million °C plasma for 101 seconds (2021).

Differentiator:

First in India to build a domestic fusion reactor, leveraging local R&D capacity while the nation is only a contributor to ITER — no indigenous project yet.

Competitor Landscape & Differentiation

3 DEEP

Global Competitors:

- Ocean Infinity (UK): advanced AUV fleets.
- WHOI (US): Woods Hole, global leader in marine robotics.
- Kongsberg (Norway): commercial AUVs for defense/industry.

Differentiator:

Integrating hyperspectral imaging with ROVs/AUVs, enabling resource mapping, marine biology, and defense applications — a niche unexplored by most global players. Positioned to align with India's ₹4,077 crore Deep Ocean Mission (2021–2026).

4 PLANCK

Global Competitors:

- IBM Quantum: 1,121-qubit "Condor" (2023).
- Google Quantum AI: achieved "quantum supremacy" (2019).
- IonQ, Rigetti: commercial quantum startups.

Differentiator:

Focus on developing error-proof, room-temperature qubits — overcoming reliance on cryogenics (~15 mK), which is today's biggest scalability barrier.

Competitor Landscape & Differentiation

5 GENETICS

Global Competitors:

- Neuralink (USA)
- Kernel (USA)
- Blue Brain Project (Switzerland)
- OpenWorm & Human Brain Project (EU)

Differentiator:

Genetics stands apart through its **hybrid neuro-quantum architecture**, enabling brain-level fidelity far beyond conventional AI or neuromorphic systems. Genetics pursues **full-scale consciousness emulation** — merging neuroscience and computation to achieve **digital immortality**.

6 BEYOND AI

Global Competitors:

- OpenAI (US): GPT models, frontier AI.
- DeepMind (UK): advanced reinforcement learning, AlphaFold.
- Anthropic, Cohere, Mistral: commercial AI labs.

Differentiator:

Unlike existing labs that rely on GPUs/TPUs, BEYOND AI will leverage PLANCK's quantum hardware to train AGI — tackling compute bottlenecks and enabling multi-domain reasoning not possible today.

DRIVE LINKS

MORE THAN THE DECK

- **FOUNDER'S INTRO :**
  https://drive.google.com/file/d/1ZL4BU29C_7TlsB2I7ea2x9CGsTkJkvC5/view?usp=sharing

- **1-MINUTE PITCH :**
  https://drive.google.com/file/d/1TglPiUc7G8T1nciIYDCU_uVcgcp2Mlcy/view?usp=sharing

- **LOGO CONCEPT :**
  https://drive.google.com/file/d/14b7P2fxcxG7bcS27MeeD9EorJBA4Zadi/view?usp=sharing

Thank YOU